Chapter 1. Distributions and Inference for Categorical Data

Binomial and multinomial likelihood inference

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1.1 Categorical response data

Motivation. Before choosing a method we must know what kind of variable we are analysing: the measurement scale of a categorical response decides which models and summaries are meaningful.

A *categorical variable* has a measurement scale consisting of a set of categories (political philosophy: liberal / moderate / conservative; a mammogram diagnosis: normal, benign, probably benign, suspicious, malignant). Such scales are pervasive in the social and biomedical sciences, but also occur in genetics, zoology, education, marketing and industrial quality control.

1.1.1 Response–explanatory variable distinction

As in regression, we distinguish a *response* (dependent) variable from *explanatory* (independent) variables. These notes concern methods for *categorical responses*; explanatory variables may be of any type.

1.1.2 Nominal–ordinal scale distinction

Definition (Nominal, ordinal, interval)

(a) A categorical variable is *nominal* if its categories have no natural ordering (religious affiliation, mode of transport, favourite music).
(b) It is *ordinal* if the categories are ordered but the distances between them are unknown (social class, patient condition good/fair/serious/critical).
(c) An *interval* variable has numerical distances between any two values (blood pressure, income); it is a *ratio* variable if ratios are also meaningful.

The classification depends on how a variable is measured: education is nominal (public/private school), ordinal (highest degree) or interval (years of schooling). In the hierarchy interval > ordinal > nominal, methods for a lower level can be used at a higher level (ignoring information) but not conversely. It is usually best to use methods appropriate to the actual scale. The book treats nominal and ordinal variables, discrete interval variables with few values, and continuous variables grouped into a few categories.

1.1.3 Continuous–discrete variable distinction

All measurement is in practice discrete; the useful distinction is between variables taking many values (often treated as continuous) and variables taking few values, which are the subject here.

1.1.4 Quantitative–qualitative variable distinction

Nominal variables are *qualitative* (categories differ in quality) and interval variables are *quantitative*. Ordinal variables sit in between: each category has more or less of the characteristic, and often an underlying continuous variable exists. Analysts exploit this by assigning *scores* to categories or by assuming a latent continuous distribution.

1.1.5 Organization of this book

The models resemble regression but replace normality by binomial, multinomial or Poisson distributions; the two central families are *logistic regression* (binary and multicategory responses) and *loglinear models* (counts). Chapters 1–3 cover distributions, inference and association measures for

one- and two-way data (largely pre-1960, non-model-based methods); Chapters 4–9 the generalized linear model family; Chapters 10–13 clustered and repeated responses; Chapters 14–16 theory and history.

1.2 Distributions for categorical data

Motivation. Inference requires a model for the random mechanism generating the counts. Three distributions play the role that the normal distribution plays for continuous responses: binomial, multinomial and Poisson.

1.2.1 Binomial distribution

Let $Y_1, \dots, Y_n$ be independent and identical binary trials with $P(Y_i = 1) = \pi$, $P(Y_i = 0) = 1 - \pi$ (*Bernoulli trials*). The total $Y = \sum_i Y_i$ has the binomial distribution $\text{bin}(n, \pi)$:

$$p(y) = \binom{n}{y} \pi^y (1-\pi)^{n-y}, \qquad y = 0, 1, \dots, n. \tag{1.1}$$

Proposition 1.1 (Binomial moments)

If $Y \sim \text{bin}(n, \pi)$ then $\mu = E(Y) = n\pi$, $\sigma^2 = \text{Var}(Y) = n\pi(1-\pi)$, and

$$\frac{E(Y-\mu)^3}{\sigma^3} = \frac{1-2\pi}{\sqrt{n\pi(1-\pi)}}.$$

In particular the distribution becomes symmetric (and, by the CLT, normal) as $n \to \infty$ for fixed π.

Proof.

Step 1. For one trial $E(Y_i) = 1 \cdot \pi + 0 \cdot (1-\pi) = \pi$ and $E(Y_i^2) = \pi$, so $\text{Var}(Y_i) = \pi - \pi^2 = \pi(1-\pi)$.

Step 2. By linearity and independence, $E(Y) = n\pi$ and $\text{Var}(Y) = n\pi(1-\pi)$.

Step 3. Third central moments of independent variables add (cumulants are additive). For one trial, $E(Y_i - \pi)^3 = (1-\pi)^3\pi + (-\pi)^3(1-\pi) = \pi(1-\pi)[(1-\pi)^2 - \pi^2] = \pi(1-\pi)(1-2\pi)$. Hence $E(Y-\mu)^3 = n\pi(1-\pi)(1-2\pi)$ and dividing by $\sigma^3 = [n\pi(1-\pi)]^{3/2}$ gives the formula, which tends to 0. ■

If trials are not independent and identical the binomial fails; e.g. sampling without replacement from a finite population gives the hypergeometric distribution (Section 3.5).

1.2.2 Multinomial distribution

Suppose each of n independent, identical trials falls in one of c categories. Let $y_{ij} = 1$ if trial i falls in category j, so $\mathbf{y}_i = (y_{i1}, \dots, y_{ic})$ has exactly one 1. Let $n_j = \sum_i y_{ij}$ and $\pi_j = P(y_{ij} = 1)$. Then $(n_1, \dots, n_c)$ is *multinomial*:

$$p(n_1, \dots, n_{c-1}) = \frac{n!}{n_1!\, n_2! \cdots n_c!}\, \pi_1^{n_1} \pi_2^{n_2} \cdots \pi_c^{n_c}. \tag{1.2}$$

Because $\sum_j n_j = n$, the distribution is $(c-1)$-dimensional; $c = 2$ gives the binomial.

Proposition 1.2 (Multinomial moments)

For the multinomial $(n, \boldsymbol{\pi})$ distribution,

$$E(n_j) = n\pi_j, \qquad \mathrm{Var}(n_j) = n\pi_j(1-\pi_j), \qquad \mathrm{Cov}(n_j, n_k) = -n\pi_j\pi_k \ (j \neq k),$$

and each n_j is marginally $\mathrm{bin}(n, \pi_j)$.

Proof. [Supplement]

Step 1. The indicator y_{ij} is Bernoulli(π_j), and n_j is a sum of n independent such indicators, so $n_j \sim \mathrm{bin}(n, \pi_j)$; the mean and variance follow from Proposition 1.1.

Step 2. For $j \neq k$, a single trial cannot fall in two categories, so $y_{ij}y_{ik} = 0$ and $\mathrm{Cov}(y_{ij}, y_{ik}) = 0 - \pi_j\pi_k$.

Step 3. Different trials are independent, so $\mathrm{Cov}(n_j, n_k) = \sum_i \mathrm{Cov}(y_{ij}, y_{ik}) = -n\pi_j\pi_k$. ■

In matrix form, $\mathrm{Cov}(\mathbf{n}) = n[\mathrm{diag}(\boldsymbol{\pi}) - \boldsymbol{\pi}\boldsymbol{\pi}^\top]$; this matrix is used repeatedly below.

1.2.3 Poisson distribution

Counts without a fixed upper limit n (e.g. weekly motorway deaths in Italy) are often modelled by the *Poisson* distribution with mean μ:

$$p(y) = \frac{e^{-\mu}\mu^y}{y!}, \qquad y = 0, 1, 2, \ldots \tag{1.3}$$

It is unimodal with mode $\lfloor \mu \rfloor$, its skewness is $E(Y-\mu)^3/\sigma^3 = 1/\sqrt{\mu}$, and it approaches normality as $\mu \to \infty$. It describes counts of events occurring randomly in time or space with independent counts in disjoint regions.

Proposition 1.3 (Poisson moments and binomial limit)

(a) If $Y \sim \mathrm{Poisson}(\mu)$ then $E(Y) = \mathrm{Var}(Y) = \mu$.
(b) If $n \to \infty$ and $\pi \to 0$ with $n\pi \to \mu$, then $\mathrm{bin}(n, \pi)$ probabilities converge to $\mathrm{Poisson}(\mu)$ probabilities.

Proof. [Supplement]

Step (a). $E(Y) = \sum_{y \geq 1} e^{-\mu}\mu^y/(y-1)! = \mu$. Similarly $E[Y(Y-1)] = \sum_{y \geq 2} e^{-\mu}\mu^y/(y-2)! = \mu^2$, so $\mathrm{Var}(Y) = \mu^2 + \mu - \mu^2 = \mu$.

Step (b). Write

$$\binom{n}{y}\pi^y(1-\pi)^{n-y} = \frac{(n\pi)^y}{y!} \cdot \frac{n(n-1)\cdots(n-y+1)}{n^y} \cdot (1-\pi)^n(1-\pi)^{-y}.$$

For fixed y the second factor tends to 1, $(1-\pi)^{-y} \to 1$, and $(1-\pi)^n = \exp\{n\log(1-\pi)\} = \exp\{-n\pi + O(n\pi^2)\} \to e^{-\mu}$. Hence the probability tends to $e^{-\mu}\mu^y/y!$. ■

For instance, if 50 million drivers each die in a given week independently with probability 0.000002, the death count is $\mathrm{bin}(5 \times 10^7, 2 \times 10^{-6})$, approximately Poisson with $\mu = 100$. Since Var $= \mu$, counts vary more when their mean is larger.

1.2.4 Overdispersion

Real counts often vary more than the binomial or Poisson allows (*overdispersion*), because the success probability or rate itself varies with unmeasured factors (driving time, seat-belt use, litter

effects in toxicology).

Proposition 1.4 (Mixing creates overdispersion)

Let Y given λ have mean $E(Y \mid \lambda)$ and variance $\mathrm{Var}(Y \mid \lambda)$, where λ is random. Then

$$E(Y) = E[E(Y \mid \lambda)], \qquad \mathrm{Var}(Y) = E[\mathrm{Var}(Y \mid \lambda)] + \mathrm{Var}[E(Y \mid \lambda)].$$

If $Y \mid \lambda \sim \text{Poisson}(\lambda)$ with $E(\lambda) = \mu$, then $E(Y) = \mu$ and $\mathrm{Var}(Y) = \mu + \mathrm{Var}(\lambda) > \mu$ whenever λ is nondegenerate.

Proof. [Supplement]

Step 1. The first identity is the tower property. For the second,

$$\begin{aligned}\mathrm{Var}(Y) &= E\big[E(Y^2 \mid \lambda)\big] - \big(E[E(Y \mid \lambda)]\big)^2 \\ &= E\big[\mathrm{Var}(Y \mid \lambda) + E(Y \mid \lambda)^2\big] - \big(E[E(Y \mid \lambda)]\big)^2 \\ &= E[\mathrm{Var}(Y \mid \lambda)] + \mathrm{Var}[E(Y \mid \lambda)].\end{aligned}$$

Step 2. In the Poisson case $E(Y \mid \lambda) = \mathrm{Var}(Y \mid \lambda) = \lambda$, which gives $E(Y) = \mu$ and $\mathrm{Var}(Y) = \mu + \mathrm{Var}(\lambda)$. ■

The same mechanism (a mixture of binomials with varying π) produces extra-binomial variation; the negative binomial (Section 4.3.4) and random-effects models (Chapters 12–13) address it.

1.2.5 Connection between Poisson and multinomial distributions

Theorem 1.5 (Poisson conditioned on the total is multinomial)

Let $Y_1, \dots, Y_c$ be independent Poisson with means $\mu_1, \dots, \mu_c$. Then $n = \sum_i Y_i \sim \text{Poisson}(\sum_i \mu_i)$, and conditional on n,

$$(Y_1, \dots, Y_c) \mid \textstyle\sum_i Y_i = n \;\sim\; \text{multinomial}(n, \{\pi_i\}), \qquad \pi_i = \mu_i / \textstyle\sum_j \mu_j.$$

Proof.

Step 1. The pgf of Y_i is $E(s^{Y_i}) = e^{\mu_i(s-1)}$; the product $e^{(\sum \mu_i)(s-1)}$ is the pgf of Poisson$(\sum \mu_i)$, so n is Poisson.

Step 2. For counts $\{n_i\}$ with $\sum n_i = n$,

$$\begin{aligned}P\Big(Y_1 = n_1, \dots, Y_c = n_c \,\Big|\, \sum Y_j = n\Big) &= \frac{\prod_i e^{-\mu_i} \mu_i^{n_i}/n_i!}{e^{-\sum_j \mu_j} (\sum_j \mu_j)^n / n!} \\ &= \frac{n!}{\prod_i n_i!} \prod_i \Big(\frac{\mu_i}{\sum_j \mu_j}\Big)^{n_i},\end{aligned}$$

using $\sum_i n_i = n$ to distribute $(\sum_j \mu_j)^n$. This is (1.2). ■

Given n the Y_i are no longer Poisson (each is at most n) nor independent. Because the Poisson and multinomial likelihoods have the same kernel in $\{\pi_i\}$, analyses under the two models usually give the same estimates.

1.3 Statistical inference for categorical data

Motivation. Once a distribution is chosen its parameters are unknown. Maximum likelihood gives estimators with known large-sample behaviour, and three likelihood-based tests (Wald, likelihood ratio, score) together with the confidence intervals obtained by inverting them form the toolkit for the whole course.

1.3.1 Likelihood functions and maximum likelihood estimation

Definition (Likelihood, ML estimate, information)

For data with a chosen distribution, the *likelihood* $\ell(\beta)$ is the probability of the observed data as a function of the parameter β, and $L(\beta) = \log \ell(\beta)$ is the *log likelihood*. The *maximum likelihood (ML) estimate* $\hat\beta$ maximizes ℓ; typically it solves the *likelihood equation* $\partial L/\partial\beta = 0$. For a parameter vector $\boldsymbol\beta$ the *information matrix* has (j,k) element

$$-E\Big(\frac{\partial^2 L(\boldsymbol\beta)}{\partial\beta_j\,\partial\beta_k}\Big). \tag{1.4}$$

Under regularity conditions the asymptotic covariance matrix of $\hat{\boldsymbol\beta}$ is the inverse information matrix; standard errors (SE) are square roots of its diagonal. Greater curvature of L means smaller SEs: the log likelihood drops quickly away from $\hat\beta$.

Lemma 1.6 (Score identities)

Let $f(y;\beta)$ be a density or pmf such that differentiation under $\int dy$ (or $\sum_y$) is allowed twice. For one observation let $u(\beta) = \partial \log f(Y;\beta)/\partial\beta$. Then

$$E_\beta[u(\beta)] = 0, \qquad \operatorname{Var}_\beta[u(\beta)] = -E_\beta\Big[\frac{\partial^2 \log f(Y;\beta)}{\partial\beta^2}\Big] =: i(\beta).$$

Proof. [Supplement]

Step 1. Differentiate $\int f(y;\beta)\,dy = 1$: $0 = \int \partial_\beta f\,dy = \int (\partial_\beta \log f)\, f\,dy = E[u]$.

Step 2. Differentiate again: $0 = \int [(\partial_\beta^2 \log f) f + (\partial_\beta \log f)^2 f]\,dy$, so $E[u^2] = -E[\partial_\beta^2 \log f]$; since $E[u] = 0$ this is $\operatorname{Var}(u)$. ∎

Theorem 1.7 (Asymptotic normality of the ML estimator)

Let $Y_1,\dots,Y_n$ be i.i.d. $f(y;\beta_0)$ with β_0 interior to the parameter space, satisfying Lemma 1.6, $0 < i(\beta_0) < \infty$, and $|\partial_\beta^3 \log f(y;\beta)| \le M(y)$ near β_0 with $E\,M(Y) < \infty$. If $\hat\beta$ is a consistent root of the likelihood equation, then

$$\sqrt{n}(\hat\beta - \beta_0) \xrightarrow{d} N\big(0,\, 1/i(\beta_0)\big).$$

For a vector $\boldsymbol\beta$ the same holds with $i(\boldsymbol\beta_0)$ the per-observation information matrix: $\sqrt{n}(\hat{\boldsymbol\beta} - \boldsymbol\beta_0) \xrightarrow{d} N(\mathbf{0}, i(\boldsymbol\beta_0)^{-1})$, and in fact $\sqrt{n}(\hat{\boldsymbol\beta} - \boldsymbol\beta_0) = i(\boldsymbol\beta_0)^{-1}\, n^{-1/2} \nabla L(\boldsymbol\beta_0) + o_p(1)$.

Proof. [Supplement]

Step 1. Taylor-expand the score $L'(\beta) = \sum_i u_i(\beta)$ about β_0:

$$0 = L'(\hat\beta) = L'(\beta_0) + (\hat\beta - \beta_0)L''(\beta_0) + \tfrac{1}{2}(\hat\beta - \beta_0)^2 L'''(\beta^*),$$

with β^* between $\hat\beta$ and β_0. Rearranging,

$$\sqrt{n}(\hat\beta-\beta_0)=\frac{n^{-1/2}L'(\beta_0)}{-n^{-1}L''(\beta_0)-\tfrac{1}{2}(\hat\beta-\beta_0)\,n^{-1}L'''(\beta^*)}.$$

Step 2. Numerator: $L'(\beta_0)$ is a sum of i.i.d. terms with mean 0 and variance $i(\beta_0)$ (Lemma 1.6), so by the CLT $n^{-1/2}L'(\beta_0)\xrightarrow{d}N(0,i(\beta_0))$.

Step 3. Denominator: by the WLLN $-n^{-1}L''(\beta_0)\xrightarrow{p}i(\beta_0)$. With probability tending to one β^* lies in the neighbourhood, where $|n^{-1}L'''(\beta^*)|\le n^{-1}\sum_i M(Y_i)=O_p(1)$; multiplied by $\hat\beta-\beta_0=o_p(1)$ it vanishes. So the denominator $\xrightarrow{p}i(\beta_0)$.

Step 4. By Slutsky's theorem $\sqrt{n}(\hat\beta-\beta_0)\xrightarrow{d}N(0,i(\beta_0))/i(\beta_0)=N(0,1/i(\beta_0))$, and the representation $\sqrt{n}(\hat\beta-\beta_0)=i(\beta_0)^{-1}n^{-1/2}L'(\beta_0)+o_p(1)$ follows from the same display. The vector case is identical with the Hessian matrix in place of L'' and matrix inversion in place of division. ■

1.3.2 Likelihood function and ML estimate for binomial parameter

The part of the likelihood involving the parameter is its *kernel*; constants such as $\binom{n}{y}$ do not affect the maximizer.

Proposition 1.8 (Binomial ML estimate and information)

For y successes in n trials, the log likelihood, score and information are

$$L(\pi)=y\log\pi+(n-y)\log(1-\pi), \tag{1.5}$$

$$\frac{\partial L}{\partial\pi}=\frac{y}{\pi}-\frac{n-y}{1-\pi}=\frac{y-n\pi}{\pi(1-\pi)}, \tag{1.6}$$

$$-E\Big(\frac{\partial^2 L}{\partial\pi^2}\Big)=\frac{n}{\pi(1-\pi)}. \tag{1.7}$$

The ML estimate is $\hat\pi=y/n$, with $E(\hat\pi)=\pi$ and $\sigma(\hat\pi)=\sqrt{\pi(1-\pi)/n}$, the inverse square root of the information.

Proof.

Step 1. (1.5) is the log of the kernel $\pi^y(1-\pi)^{n-y}$; differentiation gives (1.6), which vanishes iff $\pi=y/n$. Since $\partial^2L/\partial\pi^2=-y/\pi^2-(n-y)/(1-\pi)^2<0$, L is strictly concave and y/n is the global maximizer (for $y=0$ or n, L is monotone and the maximum is at the boundary 0 or 1, again y/n).

Step 2. Taking expectations with $E(Y)=n\pi$: $E[Y/\pi^2+(n-Y)/(1-\pi)^2]=n/\pi+n/(1-\pi)=n/[\pi(1-\pi)]$.

Step 3. $E(\hat\pi)=E(Y)/n=\pi$ and $\mathrm{Var}(\hat\pi)=n\pi(1-\pi)/n^2$, matching the inverse information. ■

1.3.3 Wald–likelihood ratio–score test triad

Consider $H_0\colon\beta=\beta_0$. Write $L_0=L(\beta_0)$ and $L_1=L(\hat\beta)$ (maximized log likelihoods under H_0 and in general).

Definition (The three likelihood-based statistics)

(a) *Wald*: $z_W=(\hat\beta-\beta_0)/\mathrm{SE}$ with SE evaluated at $\hat\beta$; chi-squared form z_W^2. For a vector, $W=(\hat{\boldsymbol\beta}-\boldsymbol\beta_0)^\top[\widehat{\mathrm{Cov}}(\hat{\boldsymbol\beta})]^{-1}(\hat{\boldsymbol\beta}-\boldsymbol\beta_0)$.

(b) *Likelihood ratio (LR)*: $\Lambda = \ell_0/\ell_1 \le 1$ and $-2\log\Lambda = -2(L_0 - L_1)$.
(c) *Score* (Rao): with $u(\beta_0) = \partial L/\partial\beta|_{\beta_0}$ and $\iota(\beta_0) = -E[\partial^2 L/\partial\beta^2]|_{\beta_0}$, $z_S = u(\beta_0)/\iota(\beta_0)^{1/2}$, chi-squared form $u(\beta_0)^2/\iota(\beta_0)$; in the vector case $\mathbf{u}^\top \iota^{-1}\mathbf{u}$ at $\boldsymbol{\beta}_0$.

The Wald test uses the behaviour of L at $\hat\beta$, the score test its slope and curvature at β_0, and the LR test both values (twice the vertical distance in Figure 1).

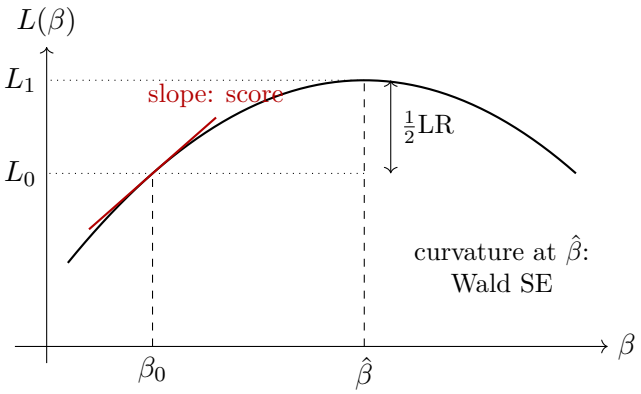


Figure 1: Information used by the three tests of $H_0\colon \beta = \beta_0$.

Lemma 1.9 (Normal quadratic forms)

(a) If $\mathbf{z} \sim N_q(\mathbf{0}, \Sigma)$ with Σ nonsingular, then $\mathbf{z}^\top\Sigma^{-1}\mathbf{z} \sim \chi^2_q$.
(b) If $\mathbf{z} \sim N_c(\mathbf{0}, P)$ with P symmetric idempotent of rank r, then $\mathbf{z}^\top\mathbf{z} \sim \chi^2_r$.

Proof. [Supplement]

Step (a). $\mathbf{w} = \Sigma^{-1/2}\mathbf{z} \sim N_q(\mathbf{0}, I)$, so $\mathbf{z}^\top\Sigma^{-1}\mathbf{z} = \sum_{k=1}^q w_k^2$ is a sum of q squared independent standard normals.

Step (b). Write $P = Q\,\mathrm{diag}(1,\dots,1,0,\dots,0)\,Q^\top$ with Q orthogonal and r ones (eigenvalues of an idempotent matrix are 0 or 1, and their number of ones is $\operatorname{rank} P$). Then $\mathbf{w} = Q^\top\mathbf{z} \sim N(\mathbf{0}, \mathrm{diag}(1,\dots,1,0,\dots,0))$: $w_1,\dots,w_r$ i.i.d. $N(0,1)$ and $w_{r+1} = \cdots = 0$ a.s. Hence $\mathbf{z}^\top\mathbf{z} = \mathbf{w}^\top\mathbf{w} = \sum_{k\le r} w_k^2 \sim \chi^2_r$. ∎

Theorem 1.10 (Asymptotic null distributions of the triad)

Under the conditions of Theorem 1.7 and $H_0\colon \boldsymbol{\beta} = \boldsymbol{\beta}_0$ ($\boldsymbol{\beta} \in \mathbb{R}^q$), the Wald, LR and score statistics each converge in distribution to χ^2_q, and the difference between any two of them converges in probability to 0. For $q = 1$, z_W and z_S are asymptotically $N(0,1)$.

Proof. [Supplement] Write $\iota_1 = i(\boldsymbol{\beta}_0)$ for the per-observation information and $\mathbf{s}_n = n^{-1/2}\nabla L(\boldsymbol{\beta}_0) \xrightarrow{d} N(\mathbf{0}, \iota_1)$ (CLT plus Lemma 1.6).

Step 1 (score). The score statistic is $\nabla L^\top (n\iota_1)^{-1}\nabla L = \mathbf{s}_n^\top \iota_1^{-1}\mathbf{s}_n \xrightarrow{d} \chi^2_q$ by the continuous mapping theorem and Lemma 1.9(a).

Step 2 (Wald). By Theorem 1.7, $\mathbf{z}_n = \sqrt{n}(\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}_0) = \iota_1^{-1}\mathbf{s}_n + o_p(1)$. The Wald statistic is $\mathbf{z}_n^\top \hat\iota_1 \mathbf{z}_n$ with $\hat\iota_1 \xrightarrow{p} \iota_1$ (consistency and continuity). Hence it equals $\mathbf{s}_n^\top\iota_1^{-1}\mathbf{s}_n + o_p(1)$, the score statistic plus $o_p(1)$.

Step 3 (LR). Expand L about $\hat{\boldsymbol{\beta}}$, where $\nabla L(\hat{\boldsymbol{\beta}}) = \mathbf{0}$:

$$L(\boldsymbol{\beta}_0) = L(\hat{\boldsymbol{\beta}}) + \tfrac{1}{2}(\boldsymbol{\beta}_0 - \hat{\boldsymbol{\beta}})^\top \nabla^2 L(\tilde{\boldsymbol{\beta}})(\boldsymbol{\beta}_0 - \hat{\boldsymbol{\beta}}),$$

with $\tilde{\boldsymbol{\beta}}$ on the segment. Hence $-2(L_0 - L_1) = \mathbf{z}_n^\top[-n^{-1}\nabla^2 L(\tilde{\boldsymbol{\beta}})]\mathbf{z}_n$, and $-n^{-1}\nabla^2 L(\tilde{\boldsymbol{\beta}}) \xrightarrow{p} \iota_1$ (WLLN plus the third-derivative bound). So $-2(L_0 - L_1) = \mathbf{z}_n^\top \iota_1 \mathbf{z}_n + o_p(1)$, again the Wald statistic plus $o_p(1)$.

Step 4. All three are $\mathbf{s}_n^\top \iota_1^{-1} \mathbf{s}_n + o_p(1) \xrightarrow{d} \chi^2_q$. For $q = 1$, $z_S = \mathbf{s}_n/\sqrt{\iota_1} \xrightarrow{d} N(0,1)$ and $z_W = z_S + o_p(1)$. ■

More generally, for a composite H_0 Wilks' theorem gives $-2\log\Lambda \xrightarrow{d} \chi^2_{\mathrm{df}}$ with df equal to the difference in dimension of the parameter spaces under $H_0 \cup H_a$ and under H_0; Theorem 1.20 below proves this in the multinomial setting. For small to moderate n the LR test is usually more reliable than the Wald test.

1.3.4 Constructing confidence intervals

A confidence interval results from *inverting* a test: the 95% interval is the set of β_0 for which the test of $H_0\colon \beta = \beta_0$ has P-value exceeding 0.05. Let z_a be the standard normal upper-a point and $\chi^2_{\mathrm{df}}(a)$ the chi-squared upper-a point.

(a) *Wald interval*: $\{\beta_0 : |\hat\beta - \beta_0|/\mathrm{SE} < z_{\alpha/2}\}$, i.e. $\hat\beta \pm z_{\alpha/2}\mathrm{SE}$.

(b) *LR interval*: $\{\beta_0 : -2[L(\beta_0) - L(\hat\beta)] < \chi^2_1(\alpha)\}$, where $\chi^2_1(\alpha) = z^2_{\alpha/2}$.

(c) *Score interval*: $\{\beta_0 : |z_S(\beta_0)| < z_{\alpha/2}\}$.

When $\hat\beta$ is normal the log likelihood is exactly parabolic and the three coincide (as in normal linear regression). With small categorical samples L can be far from parabolic and the methods disagree markedly; such disagreement signals non-normality of $\hat\beta$. The LR interval is preferable for small to moderate n; the Wald interval is the most common because it needs only $\hat\beta$ and its SE.

1.4 Statistical inference for binomial parameters

Motivation. The binomial is the simplest categorical model, yet the three large-sample methods already behave very differently near the boundary; it is the ideal test case for comparing Wald, score, LR and exact inference.

1.4.1 Tests about a binomial parameter

Proposition 1.11 (Binomial test statistics)

For $H_0\colon \pi = \pi_0$ with $\hat\pi = y/n$:

$$z_W = \frac{\hat\pi - \pi_0}{\sqrt{\hat\pi(1-\hat\pi)/n}}, \tag{1.8}$$

$$z_S = \frac{\hat\pi - \pi_0}{\sqrt{\pi_0(1-\pi_0)/n}}, \tag{1.9}$$

$$-2(L_0 - L_1) = 2\sum \text{observed} \log\frac{\text{observed}}{\text{fitted}} = 2\Big[y\log\frac{y}{n\pi_0} + (n-y)\log\frac{n-y}{n-n\pi_0}\Big]. \tag{1.10}$$

Each is asymptotically $N(0,1)$ or χ^2_1 under H_0.

Proof.

Step 1. The Wald statistic uses $\mathrm{SE} = \sqrt{\hat\pi(1-\hat\pi)/n}$ (Proposition 1.8 with $\pi = \hat\pi$).

Step 2. Score: by (1.6)–(1.7),

$$\frac{u(\pi_0)}{\iota(\pi_0)^{1/2}} = \frac{y - n\pi_0}{\pi_0(1-\pi_0)} \cdot \sqrt{\frac{\pi_0(1-\pi_0)}{n}} = \frac{y - n\pi_0}{\sqrt{n\pi_0(1-\pi_0)}} = \frac{\hat\pi - \pi_0}{\sqrt{\pi_0(1-\pi_0)/n}}.$$

Step 3. LR: $L_0 = y\log\pi_0 + (n-y)\log(1-\pi_0)$ and $L_1 = y\log\hat\pi + (n-y)\log(1-\hat\pi)$, so $-2(L_0 - L_1) = 2[y\log(\hat\pi/\pi_0) + (n-y)\log((1-\hat\pi)/(1-\pi_0))]$, which is (1.10) after multiplying numerator and denominator by n. The asymptotics are Theorem 1.10 with $q = 1$. ■

The score statistic uses the actual null SE rather than an estimate, and its null distribution is closer to $N(0,1)$. Form (1.10) (observed vs. fitted counts) recurs for Poisson and multinomial tests.

1.4.2 Confidence intervals for a binomial parameter

Inverting the Wald test gives the Laplace interval

$$\hat\pi \pm z_{\alpha/2}\sqrt{\hat\pi(1-\hat\pi)/n}, \tag{1.11}$$

which performs poorly unless n is very large: its actual coverage is usually below nominal, much below for π near 0 or 1. (Adding $z_{\alpha/2}^2/2 \approx 2$ observations of each type before applying (1.11) improves it greatly.)

Proposition 1.12 (Wilson score interval)

The score interval $\{\pi_0 : |\hat\pi - \pi_0| < z\sqrt{\pi_0(1-\pi_0)/n}\}$, $z = z_{\alpha/2}$, has endpoints

$$\hat\pi\Big(\frac{n}{n+z^2}\Big) + \frac{1}{2}\Big(\frac{z^2}{n+z^2}\Big) \;\pm\; z\sqrt{\frac{1}{n+z^2}\Big[\hat\pi(1-\hat\pi)\Big(\frac{n}{n+z^2}\Big) + \frac{1}{2}\cdot\frac{1}{2}\Big(\frac{z^2}{n+z^2}\Big)\Big]}.$$

Its midpoint is $(y + z^2/2)/(n+z^2)$, the sample proportion after adding $z^2/2$ successes and $z^2/2$ failures.

Proof. [Supplement]

Step 1. The endpoints solve $(\hat\pi - \pi_0)^2 = z^2\pi_0(1-\pi_0)/n$, i.e.

$$\Big(1 + \frac{z^2}{n}\Big)\pi_0^2 - \Big(2\hat\pi + \frac{z^2}{n}\Big)\pi_0 + \hat\pi^2 = 0.$$

The quadratic is negative between its roots, which is exactly the set where $|z_S| < z$.

Step 2. Multiplying by n and using the quadratic formula, the roots are

$$\pi_0 = \frac{n\hat\pi + z^2/2 \pm z\sqrt{n\hat\pi(1-\hat\pi) + z^2/4}}{n+z^2}.$$

Step 3. The centre is $(n\hat\pi + z^2/2)/(n+z^2) = \hat\pi\,\frac{n}{n+z^2} + \frac{1}{2}\,\frac{z^2}{n+z^2}$. For the half-width, $z\sqrt{n\hat\pi(1-\hat\pi) + z^2/4}/(n+z^2) = z\sqrt{[\hat\pi(1-\hat\pi)\frac{n}{n+z^2} + \frac{1}{4}\frac{z^2}{n+z^2}]/(n+z^2)}$, which is the stated form. ■

The squared coefficient of z is a weighted average of the variance of a sample proportion at $\pi = \hat\pi$ and at $\pi = 1/2$, with sample size $n + z^2$. The Wilson interval performs much better than the Wald interval. The *LR interval* is the set of π_0 for which $-2[L(\pi_0) - L(\hat\pi)] < \chi_1^2(\alpha)$.

1.4.3 Proportion of vegetarians example

Example 1.13 (Proportion of vegetarians)

Of $n = 25$ students in a statistics class, $y = 0$ said they were vegetarian, so $\hat{\pi} = 0$.
(a) Wald: $0 \pm 1.96\sqrt{0 \cdot 1/25} = (0, 0)$, a useless answer at the boundary.
(b) Score: $(0.0,\ 0.133)$. E.g. for $\pi_0 = 0.5$, $z_S = (0 - 0.5)/\sqrt{0.5 \cdot 0.5/25} = -5.0$ (rejected), while for $\pi_0 = 0.10$, $z_S = (0 - 0.10)/\sqrt{0.10 \cdot 0.90/25} = -1.67$ (inside the interval).
(c) LR: $L(\pi) = 25\log(1 - \pi)$ and $L(\hat{\pi}) = 0$, so the interval is $\{\pi_0 : -50\log(1 - \pi_0) \le 3.84\}$; the upper bound is $1 - \exp(-3.84/50) = 0.074$, giving $(0.0,\ 0.074)$.
The three large-sample methods differ greatly: for π near 0 and small n, $\hat{\pi}$ is highly skewed.

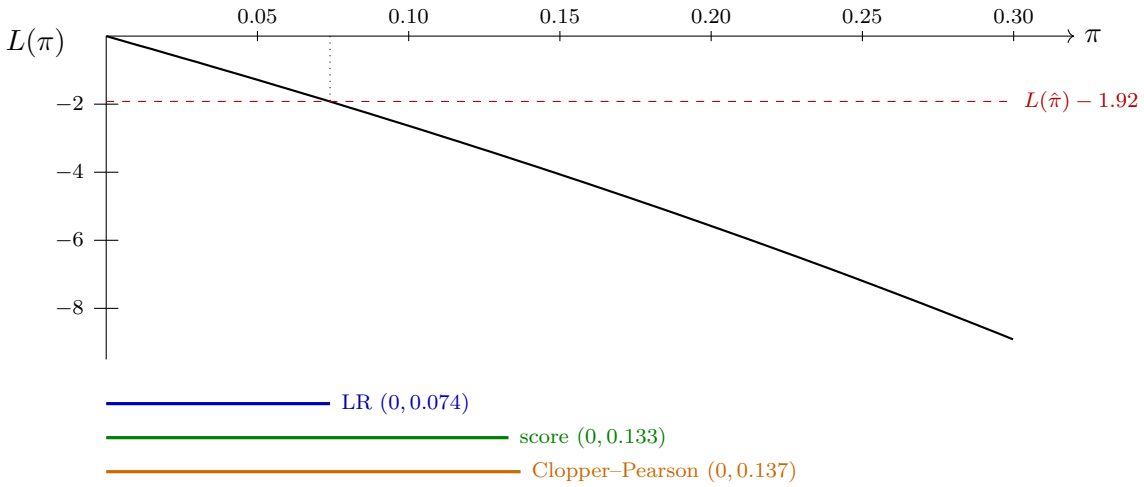


Figure 2: Binomial log likelihood for $y = 0$, $n = 25$; the LR interval is where L is within 3.84/2 of its maximum.

1.4.4 Exact small-sample inference*

Tests and intervals can use the binomial distribution directly. For $H_0\colon \pi = 0.5$ vs. $H_a\colon \pi \neq 0.5$ with $y = 0$, $n = 25$, the exact P-value for $|z_S| \ge 5$ is $P(Y = 0) + P(Y = 25) = 2(0.5)^{25} = 0.00000006$.

The *Clopper–Pearson* interval inverts two one-sided exact tests (the *tail method*): its endpoints solve

$$\sum_{k=y}^{n} \binom{n}{k} \pi_L^k (1 - \pi_L)^{n-k} = \frac{\alpha}{2}, \qquad \sum_{k=0}^{y} \binom{n}{k} \pi_U^k (1 - \pi_U)^{n-k} = \frac{\alpha}{2},$$

with $\pi_L = 0$ if $y = 0$ and $\pi_U = 1$ if $y = n$.

Theorem 1.14 (Clopper–Pearson interval: beta form and conservatism)

(a) For $1 \le y \le n$, $P_\pi(Y \ge y) = P(B \le \pi)$ where $B \sim \text{Beta}(y,\ n - y + 1)$. Consequently, for $1 \le y \le n - 1$,

$$\pi_L = \text{Beta}_{\alpha/2}(y,\ n - y + 1), \qquad \pi_U = \text{Beta}_{1-\alpha/2}(y + 1,\ n - y),$$

where $\text{Beta}_q(a, b)$ is the q quantile. (Equivalently, via $F = \frac{b}{a}\frac{B}{1-B} \sim F_{2a,2b}$, the endpoints can be written with F quantiles.)
(b) For every $\pi \in [0, 1]$, $P_\pi(\pi_L(Y) \le \pi \le \pi_U(Y)) \ge 1 - \alpha$.

Proof. [Supplement]

Step 1. Let $U_1, \ldots, U_n$ be i.i.d. uniform on $(0, 1)$ and $Y = \#\{i : U_i \le \pi\} \sim \text{bin}(n, \pi)$. Then $Y \ge y$ iff the yth order statistic satisfies $U_{(y)} \le \pi$. The density of $U_{(y)}$ is $\frac{n!}{(y-1)!(n-y)!} u^{y-1}(1 - u)^{n-y}$, the

Beta$(y, n-y+1)$ density. This proves the identity, and shows $P_\pi(Y \geq y)$ is continuous and strictly increasing in π.

Step 2. Hence $P_{\pi_L}(Y \geq y) = \alpha/2$ means π_L is the $\alpha/2$ quantile of Beta$(y, n-y+1)$. Also $P_\pi(Y \leq y) = 1 - P_\pi(Y \geq y+1) = \alpha/2$ means $P(\text{Beta}(y+1, n-y) \leq \pi_U) = 1-\alpha/2$.

Step 3. For (b), fix π. By Step 1, $\pi < \pi_L(y)$ iff $P_\pi(Y \geq y) < \alpha/2$. Since $P_\pi(Y \geq y)$ decreases in y, the set of such y is an upper set $\{y \geq k\}$ with $P_\pi(Y \geq k) < \alpha/2$; hence $P_\pi(\pi < \pi_L(Y)) < \alpha/2$. Symmetrically $P_\pi(\pi > \pi_U(Y)) < \alpha/2$, and the non-coverage probability is at most α. ■

For $y = 0$, $n = 25$ the Clopper–Pearson 95% interval is $(0.0,\ 0.137)$. Because of discreteness the actual coverage is at least, and can be much greater than, the nominal level; similarly a fixed test size such as 0.05 is usually not attainable. Figure 1.3 of the book shows, for $n = 25$, coverage too low for the Wald interval, too high for Clopper–Pearson, and close to nominal (except near 0 or 1) for the score interval. Coverage at each π is a sum of between 0 and 26 binomial probabilities and jumps as π enters or leaves one of the 26 possible intervals.

Shorter exact intervals result from inverting a single two-sided test. In *Blaker's* method the P-value is the smaller one-tailed probability plus the largest attainable probability in the other tail not exceeding it; for the vegetarian data it gives $(0.0,\ 0.128)$.

1.4.5 Inference based on the mid-P-value*

Definition (Mid-P-value)

For a test statistic T with observed value t, where large T contradicts H_0,

$$\text{mid-}P = \tfrac{1}{2}P(T = t) + P(T > t),$$

computed under H_0: the ordinary P-value minus half the probability of the observed result.

Proposition 1.15 (Properties of the mid-P-value)

Let T be discrete under H_0.
(a) The two one-sided mid-P-values sum to 1.
(b) $E_{H_0}(\text{mid-}P) = 1/2$, whereas the ordinary P-value has null mean $\frac{1}{2} + \frac{1}{2}\sum_t P(T=t)^2 > \frac{1}{2}$.

Proof. [Supplement] Let $p_t = P_{H_0}(T = t)$.

Step (a). $[\frac{1}{2}p_t + P(T > t)] + [\frac{1}{2}p_t + P(T < t)] = P(T > t) + p_t + P(T < t) = 1$.

Step (b).

$$E(\text{mid-}P) = \sum_t p_t\Big(\sum_{s>t} p_s + \tfrac{1}{2}p_t\Big) = \sum_{s>t} p_t p_s + \tfrac{1}{2}\sum_t p_t^2 = \tfrac{1}{2}\Big(\sum_t p_t\Big)^2 = \tfrac{1}{2},$$

since $(\sum p_t)^2 = \sum_t p_t^2 + 2\sum_{s>t} p_t p_s$. For the ordinary P-value add $\frac{1}{2}\sum_t p_t^2$. ■

So the mid-P-value behaves more like a continuous P-value (null distribution closer to uniform). It does not guarantee type I error $\leq \alpha$, but usually performs well and is less conservative than the exact test. For the vegetarian data the mid-P-value for $H_0\colon \pi = 0.5$ is 0.00000003, and the Clopper–Pearson-type interval using mid-P is $(0.000,\ 0.113)$, versus $(0.000,\ 0.137)$. It is a sensible compromise between overly conservative inference and artificial randomization, recommended for highly discrete distributions.

1.5 Statistical inference for multinomial parameters

Motivation. With $c > 2$ categories the natural questions are goodness of fit to a specified distribution, or to a distribution with a few unknown parameters; this leads to Pearson's X^2 and the likelihood-ratio G^2, the two workhorse statistics of categorical data analysis.

1.5.1 Estimation of multinomial parameters

Proposition 1.16 (Multinomial ML estimates)

The multinomial kernel

$$\prod_j \pi_j^{n_j}, \qquad \pi_j \geq 0, \quad \sum_j \pi_j = 1, \tag{1.12}$$

is maximized uniquely by the sample proportions $\hat{\pi}_j = p_j = n_j/n$.

Proof.

Step 1 (likelihood equations). Treat $L(\boldsymbol{\pi}) = \sum_j n_j \log \pi_j$ as a function of $\pi_1, \ldots, \pi_{c-1}$ with $\pi_c = 1 - \sum_{j<c} \pi_j$, so $\partial \pi_c / \partial \pi_j = -1$. Then

$$\frac{\partial L}{\partial \pi_j} = \frac{n_j}{\pi_j} - \frac{n_c}{\pi_c} = 0, \qquad j = 1, \ldots, c-1,$$

so $\hat{\pi}_j / \hat{\pi}_c = n_j / n_c$ for all j. Summing, $1/\hat{\pi}_c = n/n_c$, hence $\hat{\pi}_c = n_c/n$ and $\hat{\pi}_j = n_j/n$.

Step 2 (global maximum). [Supplement] For any admissible $\boldsymbol{\pi}$, with $p_j = n_j/n$ and the sum over j with $n_j > 0$, Jensen's inequality for the concave log gives

$$L(\boldsymbol{\pi}) - L(\mathbf{p}) = n \sum_j p_j \log \frac{\pi_j}{p_j} \leq n \log \sum_j p_j \frac{\pi_j}{p_j} \leq n \log 1 = 0,$$

with equality iff $\pi_j = p_j$ for all j. ∎

1.5.2 Pearson statistic for testing a specified multinomial

Karl Pearson's 1900 test (originally for a Monte Carlo roulette wheel) was among the first inferential methods. For $H_0 \colon \pi_j = \pi_{j0}$, $j = 1, \ldots, c$, the *expected frequencies* are $\mu_j = n\pi_{j0}$ and

$$X^2 = \sum_j \frac{(n_j - \mu_j)^2}{\mu_j}. \tag{1.13}$$

The exact P-value is the sum of null multinomial probabilities of all count vectors with $X^2 \geq X_o^2$ (observed value); for large n, $X^2 \approx \chi^2_{c-1}$ (Theorem 1.18).

1.5.3 Example: testing Mendel's theories

Example 1.17 (Mendel's peas)

Crossing pure yellow with pure green pea strains, Mendel predicted 75% yellow and 25% green second-generation seeds. One experiment gave $n = 8023$ seeds: $n_1 = 6022$ yellow, $n_2 = 2001$ green. Under $H_0 \colon \pi_1 = 0.75$, $\mu_1 = 6017.25$ and $\mu_2 = 2005.75$, and $X^2 = 0.015$ (df $= 1$), $P = 0.90$: no contradiction.

Fisher (1936) combined Mendel's experiments using the reproductive property of independent chi-squared statistics ($\sum X_i^2 \sim \chi^2_{\sum \nu_i}$) and obtained $X^2 = 42$ with df $= 84$. Since χ^2_{84} has mean 84 and standard deviation $\sqrt{2 \cdot 84} = 13.0$, $P(\chi^2_{84} \le 42)$ is tiny: the right-tail probability is 0.99996. The fit was *too good*, suggesting the data had been adjusted to agree with expectations.

Goodness-of-fit tests can thus reveal fits better than chance would allow.

1.5.4 Chi-squared theoretical justification*

Theorem 1.18 (Asymptotic null distribution of Pearson's X^2)

Under $H_0\colon \boldsymbol{\pi} = \boldsymbol{\pi}_0$ with all $\pi_{j0} > 0$, $X^2 \xrightarrow{d} \chi^2_{c-1}$ as $n \to \infty$. Moreover X^2 is the score statistic for H_0.

Proof. [Supplement] (Tools: multivariate CLT, Lemma 1.9.) Let $\hat{\boldsymbol{\pi}} = (p_1, \dots, p_{c-1})^\top$, $\boldsymbol{\pi}_0 = (\pi_{10}, \dots, \pi_{c-1,0})^\top$ and $\Sigma_0 = \operatorname{diag}(\boldsymbol{\pi}_0) - \boldsymbol{\pi}_0 \boldsymbol{\pi}_0^\top$.

Step 1 (CLT). $\hat{\boldsymbol{\pi}}$ is the mean of the i.i.d. vectors $(y_{i1}, \dots, y_{i,c-1})$, whose covariance is Σ_0 (Proposition 1.2). Hence $\sqrt{n}(\hat{\boldsymbol{\pi}} - \boldsymbol{\pi}_0) \xrightarrow{d} N_{c-1}(\mathbf{0}, \Sigma_0)$.

Step 2 (inverse). $\Sigma_0^{-1} = \operatorname{diag}(1/\pi_{j0}) + \mathbf{1}\mathbf{1}^\top/\pi_{c0}$. Indeed, using $\boldsymbol{\pi}_0^\top \mathbf{1} = 1 - \pi_{c0}$,

$$\Sigma_0\Big[\operatorname{diag}(1/\pi_{j0}) + \tfrac{\mathbf{1}\mathbf{1}^\top}{\pi_{c0}}\Big] = I + \frac{\boldsymbol{\pi}_0 \mathbf{1}^\top}{\pi_{c0}} - \boldsymbol{\pi}_0 \mathbf{1}^\top - \frac{(1-\pi_{c0})\boldsymbol{\pi}_0 \mathbf{1}^\top}{\pi_{c0}}$$
$$= I + \boldsymbol{\pi}_0 \mathbf{1}^\top \frac{1 - \pi_{c0} - (1-\pi_{c0})}{\pi_{c0}} = I.$$

Step 3 (algebra). Put $d_j = p_j - \pi_{j0}$, $j = 1, \dots, c$; then $d_c = -\sum_{j<c} d_j$. So

$$n\, \mathbf{d}^\top \Sigma_0^{-1} \mathbf{d} = n\Big[\sum_{j<c} \frac{d_j^2}{\pi_{j0}} + \frac{(\sum_{j<c} d_j)^2}{\pi_{c0}}\Big] = n \sum_{j=1}^{c} \frac{d_j^2}{\pi_{j0}} = \sum_{j=1}^{c} \frac{(n_j - n\pi_{j0})^2}{n\pi_{j0}} = X^2. \tag{1.14}$$

Step 4 (limit). By the continuous mapping theorem and Lemma 1.9(a), $X^2 \xrightarrow{d} \mathbf{z}^\top \Sigma_0^{-1} \mathbf{z} \sim \chi^2_{c-1}$ with $\mathbf{z} \sim N(\mathbf{0}, \Sigma_0)$.

Step 5 (score test). From Step 1 of Proposition 1.16, the score has components $u_j = n_j/\pi_j - n_c/\pi_c = n(d_j/\pi_j - d_c/\pi_c)$ at $\boldsymbol{\pi}_0$, i.e. $\mathbf{u} = n\Sigma_0^{-1}\mathbf{d}$ (by Step 2, $(\Sigma_0^{-1}\mathbf{d})_j = d_j/\pi_{j0} + \sum_{k<c} d_k/\pi_{c0} = d_j/\pi_{j0} - d_c/\pi_{c0}$). The information is $n\Sigma_0^{-1}$ (the inverse of $\operatorname{Cov}(\hat{\boldsymbol{\pi}}) = \Sigma_0/n$, or directly by differentiating $\mathbf{u}$). So the score statistic is $\mathbf{u}^\top (n\Sigma_0^{-1})^{-1} \mathbf{u} = n \mathbf{d}^\top \Sigma_0^{-1} \mathbf{d} = X^2$. ∎

Fisher's heuristic

If $n_1, \dots, n_c$ were independent Poisson with large means μ_j, the $z_j = (n_j - \mu_j)/\sqrt{\mu_j}$ would be approximately $N(0,1)$ and $\sum z_j^2 = X^2 \approx \chi^2_c$. Conditioning on $\sum n_j = n$ (Theorem 1.5) imposes one linear constraint $\sum(n_j - \mu_j) = 0$ and loses one degree of freedom. For $c = 2$, $X^2 = z_S^2$ with z_S from (1.9): for Mendel's data $\hat{\pi}_1 = 6022/8023$, $z_S = 0.123$ and $X^2 = 0.123^2 = 0.015$.

1.5.5 Likelihood-ratio chi-squared

Under H_0 the kernel (1.12) is maximized at $\boldsymbol{\pi}_0$, in general at $p_j = n_j/n$. Hence $\Lambda = \prod_j (\pi_{j0}/p_j)^{n_j}$ and

$$G^2 = -2\log\Lambda = 2\sum_j n_j \log\frac{n_j}{n\pi_{j0}}, \tag{1.15}$$

of the form (1.10). The general parameter space has dimension $c-1$, the null space dimension 0, so $G^2 \xrightarrow{d} \chi^2_{c-1}$ under H_0 (Theorem 1.10 with $q = c-1$).

Theorem 1.19 (Asymptotic equivalence of X^2 and G^2)

Let μ_j be (possibly estimated) expected frequencies with $\sum_j \mu_j = n$ and $(n_j - \mu_j)/\mu_j = O_p(n^{-1/2})$ for each j, as holds under H_0. Then $G^2 - X^2 \xrightarrow{p} 0$; in particular $G^2 \xrightarrow{d} \chi^2_{c-1}$ under $H_0\colon \boldsymbol{\pi} = \boldsymbol{\pi}_0$.

Proof. [Supplement]

Step 1. Write $n_j = \mu_j(1+x_j)$ with $x_j = O_p(n^{-1/2})$. Since $(1+x)\log(1+x) = x + x^2/2 + R(x)$ with $|R(x)| \le C|x|^3$ for $|x| \le 1/2$,

$$G^2 = 2\sum_j \mu_j(1+x_j)\log(1+x_j) = 2\sum_j \mu_j x_j + \sum_j \mu_j x_j^2 + 2\sum_j \mu_j R(x_j).$$

Step 2. $\sum_j \mu_j x_j = \sum_j (n_j - \mu_j) = 0$, and $\sum_j \mu_j x_j^2 = \sum_j (n_j-\mu_j)^2/\mu_j = X^2$.

Step 3. $\mu_j = O(n)$ and $R(x_j) = O_p(n^{-3/2})$, so the remainder is $O_p(n^{-1/2}) \xrightarrow{p} 0$. Hence $G^2 = X^2 + o_p(1)$, and Slutsky with Theorem 1.18 gives the limit. ∎

When H_0 is false both statistics grow proportionally to n but need not be close. For fixed c, X^2 usually approaches chi-squared faster than G^2; the approximation is often poor for G^2 when $n/c < 5$, while for X^2 it can be adequate with n/c as small as 1 if the table does not mix very small and moderately large expected frequencies. Exact distributions can also be computed from the multinomial.

1.5.6 Testing with estimated expected frequencies

Often $\pi_{j0} = \pi_j(\boldsymbol{\theta})$ depends on unknown $\boldsymbol{\theta}$ with $\dim\boldsymbol{\theta} = p$; the ML estimate $\hat{\boldsymbol{\theta}}$ gives $\hat\mu_j = n\pi_j(\hat{\boldsymbol{\theta}})$, and $X^2 = \sum (n_j - \hat\mu_j)^2/\hat\mu_j$. Pearson did not realize that estimating $\boldsymbol{\theta}$ changes the degrees of freedom.

Theorem 1.20 (Degrees of freedom with estimated parameters)

Assume $\boldsymbol{\pi}(\boldsymbol{\theta})$ has continuous second derivatives, $\pi_j(\boldsymbol{\theta}_0) > 0$, the $c \times p$ Jacobian $\partial\boldsymbol{\pi}/\partial\boldsymbol{\theta}^\top$ has rank p at $\boldsymbol{\theta}_0$, and $\hat{\boldsymbol{\theta}}$ is a consistent ML estimate. If $H_0\colon \boldsymbol{\pi} = \boldsymbol{\pi}(\boldsymbol{\theta})$ holds, then

$$X^2 = \sum_j \frac{(n_j - \hat\mu_j)^2}{\hat\mu_j} \xrightarrow{d} \chi^2_{c-1-p}, \qquad \text{and } G^2 - X^2 \xrightarrow{p} 0.$$

Proof. [Supplement] (Tools: multivariate CLT, Theorem 1.7, Slutsky, Lemma 1.9(b).) Let $\boldsymbol{\pi}_0 = \boldsymbol{\pi}(\boldsymbol{\theta}_0)$ (all c components), $D = \operatorname{diag}(\boldsymbol{\pi}_0)$, $\mathbf{s} = (\sqrt{\pi_{10}}, \dots, \sqrt{\pi_{c0}})^\top$ (a unit vector), and $A = D^{-1/2}\,\partial\boldsymbol{\pi}/\partial\boldsymbol{\theta}^\top$ at $\boldsymbol{\theta}_0$ ($c \times p$, rank p).

Step 1. Since $\sum_j \pi_j(\boldsymbol{\theta}) = 1$, $\mathbf{s}^\top A = \sum_j \partial\pi_j/\partial\boldsymbol{\theta}^\top = \mathbf{0}$.

Step 2. With $\mathbf{p} = (p_1, \dots, p_c)^\top$, $\operatorname{Cov}(\sqrt{n}\,\mathbf{p}) = D - \boldsymbol{\pi}_0\boldsymbol{\pi}_0^\top$, so by the CLT

$$\mathbf{z}_n := \sqrt{n}\,D^{-1/2}(\mathbf{p} - \boldsymbol{\pi}_0) \xrightarrow{d} \mathbf{z} \sim N_c(\mathbf{0},\, I - \mathbf{s}\mathbf{s}^\top).$$

Step 3. The log likelihood is $L(\boldsymbol{\theta}) = n\sum_j p_j \log \pi_j(\boldsymbol{\theta})$, with score at $\boldsymbol{\theta}_0$

$$\nabla L = n\sum_j p_j \frac{\partial \pi_j/\partial \boldsymbol{\theta}}{\pi_j} = n\, A^\top D^{-1/2}\mathbf{p} = n\, A^\top D^{-1/2}(\mathbf{p}-\boldsymbol{\pi}_0) = \sqrt{n}\, A^\top \mathbf{z}_n,$$

using $A^\top D^{-1/2}\boldsymbol{\pi}_0 = A^\top \mathbf{s} = \mathbf{0}$. The per-observation information is $\sum_j (\partial\pi_j/\partial\boldsymbol{\theta})(\partial\pi_j/\partial\boldsymbol{\theta})^\top/\pi_j = A^\top A$ (nonsingular). By Theorem 1.7, $\sqrt{n}(\hat{\boldsymbol{\theta}} - \boldsymbol{\theta}_0) = (A^\top A)^{-1}A^\top \mathbf{z}_n + o_p(1)$.

Step 4. By the delta method (a first-order Taylor expansion of $\boldsymbol{\pi}$), $\sqrt{n}\, D^{-1/2}(\boldsymbol{\pi}(\hat{\boldsymbol{\theta}}) - \boldsymbol{\pi}_0) = A\sqrt{n}(\hat{\boldsymbol{\theta}} - \boldsymbol{\theta}_0) + o_p(1)$. Therefore, with $H = A(A^\top A)^{-1}A^\top$,

$$\mathbf{e}_n := \sqrt{n}\, D^{-1/2}(\mathbf{p} - \boldsymbol{\pi}(\hat{\boldsymbol{\theta}})) = (I - H)\mathbf{z}_n + o_p(1) \xrightarrow{d} (I-H)\mathbf{z}.$$

Step 5. $X^2 = \sum_j n(p_j - \pi_j(\hat{\boldsymbol{\theta}}))^2/\pi_j(\hat{\boldsymbol{\theta}})$ differs from $\mathbf{e}_n^\top \mathbf{e}_n$ only by replacing $\pi_j(\hat{\boldsymbol{\theta}})$ by π_{j0} in the denominators, a factor $\xrightarrow{p} 1$; so $X^2 = \mathbf{e}_n^\top \mathbf{e}_n + o_p(1)$.

Step 6. The limit $(I-H)\mathbf{z}$ is normal with covariance $(I-H)(I-\mathbf{s}\mathbf{s}^\top)(I-H) = I - H - \mathbf{s}\mathbf{s}^\top$, because $H\mathbf{s} = A(A^\top A)^{-1}A^\top \mathbf{s} = \mathbf{0}$. Here H and $\mathbf{s}\mathbf{s}^\top$ are orthogonal projections onto orthogonal subspaces of dimensions p and 1, so $P = I - H - \mathbf{s}\mathbf{s}^\top$ is a symmetric idempotent matrix of rank $c - p - 1$. By Lemma 1.9(b), $X^2 \xrightarrow{d} \chi^2_{c-1-p}$.

Step 7. Step 4 shows $(n_j - \hat{\mu}_j)/\hat{\mu}_j = O_p(n^{-1/2})$, and $\sum_j \hat{\mu}_j = n$; so Theorem 1.19 gives $G^2 - X^2 \xrightarrow{p} 0$. ■

Example 1.21 (Pneumonia infections in calves)

156 dairy calves were classified by primary pneumonia infection within 60 days of birth and, for infected calves, secondary infection within two weeks after recovery. "No primary, yes secondary" is impossible (a *structural zero*). Estimated expected frequencies are in parentheses.

	Secondary infection	
Primary infection	Yes	No
Yes	30 (38.1)	63 (39.0)
No	–	63 (78.9)

The hypothesis that $P(\text{primary}) = \pi$ equals $P(\text{secondary} \mid \text{primary})$ gives trinomial probabilities $(\pi^2,\ \pi(1-\pi),\ 1-\pi)$ for (yes,yes), (yes,no), (no,no). The kernel is

$$(\pi^2)^{n_{11}}[\pi(1-\pi)]^{n_{12}}(1-\pi)^{n_{22}} = \pi^{2n_{11}+n_{12}}(1-\pi)^{n_{12}+n_{22}},$$

so the likelihood equation $\frac{2n_{11}+n_{12}}{\pi} - \frac{n_{12}+n_{22}}{1-\pi} = 0$ gives

$$\hat{\pi} = \frac{2n_{11}+n_{12}}{2n_{11}+2n_{12}+n_{22}} = \frac{123}{249} = 0.494.$$

Then $\hat{\mu}_{11} = n\hat{\pi}^2 = 38.1$, $\hat{\mu}_{12} = n(\hat{\pi} - \hat{\pi}^2) = 39.0$, $\hat{\mu}_{22} = n(1-\hat{\pi}) = 78.9$, and $X^2 = 19.7$ with $\text{df} = 3 - 1 - 1 = 1$ ($P < 0.00001$). Far more calves had a primary but no secondary infection than predicted: the primary infection seems to have an immunizing effect.

1.6 R lab

Motivation. Each inference in this chapter is one line of base R; reproducing the examples shows how the Wald, score, LR and exact intervals and the X^2/G^2 tests are computed in practice.

Binomial intervals for the vegetarian data. `prop.test` without continuity correction gives the Wilson (score) interval; `binom.test` gives the exact test and Clopper–Pearson interval; the LR interval is solved with `uniroot`.

```
y <- 0; n <- 25
prop.test(y, n, p = 0.5, correct = FALSE)$conf.int   # score: (0, 0.133)
binom.test(y, n, p = 0.5)                            # exact P, CP: (0, 0.137)
LR <- function(p0) -2 * (y*log(p0) + (n-y)*log(1-p0))  # L(pihat) = 0 here
uniroot(function(p) LR(p) - qchisq(0.95, 1), c(1e-6, 0.5))$root  # 0.074
```

Mid-P Clopper–Pearson bound. The upper limit solves $\frac{1}{2}P(Y = 0) + P(Y < 0) = 0.025$ when $y = 0$.

```
uniroot(function(p) 0.5 * dbinom(0, 25, p) - 0.025, c(1e-6, 0.5))$root  # 0.113
```

Pearson and LR goodness of fit (Mendel).

```
obs <- c(6022, 2001); p0 <- c(0.75, 0.25)
chisq.test(obs, p = p0)                        # X^2 = 0.015, P = 0.90
mu <- sum(obs) * p0
G2 <- 2 * sum(obs * log(obs / mu)); 1 - pchisq(G2, df = 1)
```

Estimated expected frequencies (calves). `chisq.test` would use df $= c - 1$, so compute X^2 and the corrected df by hand.

```
n11 <- 30; n12 <- 63; n22 <- 63; N <- n11 + n12 + n22
pih <- (2*n11 + n12) / (2*n11 + 2*n12 + n22)          # 0.494
mu  <- N * c(pih^2, pih*(1 - pih), 1 - pih)           # 38.1 39.0 78.9
X2  <- sum((c(n11, n12, n22) - mu)^2 / mu)            # 19.7
pchisq(X2, df = 3 - 1 - 1, lower.tail = FALSE)
```